

Multiple Choice (10 marks)

1. If $\mathbf{r}(t) = (6t+2)\mathbf{i} + 5t^2\mathbf{j} - 8t\mathbf{k}$, find the Binormal vector as $[\mathbf{x}, \mathbf{y}, \mathbf{z}]$. What are x , y , z ? Return them as a list.

0.7, 0.0, 0.7

0.6, 0.0, 0.8

0.0, 0.8, 0.6

0.0, 0.7, 0.7

0.8, 0.6, 0.0

2. Use the Trapezoidal Rule with to approximate $\int_0^\pi \sin^2(x)dx$. Return the approximated demical value.

(a) 1.414214

(b) 1.000000

(c) 0.785398

(d) 1.570796

(e) 1.234567

3. What is the smallest whole number that has a remainder of 1 when divided by 4, a remainder of 1 when divided by 3, and a remainder of 2 when divided by 5?

(a) 72

(b) 76

(c) 67

(d) 64

(e) 57

4. Find all zeros in the indicated finite field of the given polynomial with coefficients in that field. $x^5 + 3x^3 + x^2 + 2x$ in $\mathbb{Z}_5$

(a) 1,2

(b) 0,1,2,3

(c) 1

(d) 3,4

(e) 0,1,2

5. Each story of a building is 14.3 feet tall. If the height of the building is 185.9 feet, how many floors does the building have?

- (a) 10
- (b) 15
- (c) 17
- (d) 9
- (e) 11

True / False (10 marks)

Answer True or False

6. True or False: The bakers can make a total of 41 apple pies using 270 apples from 15 boxes.
7. True or False: The field of real numbers $\mathbb{R}$ is indeed a splitting field for polynomials with real coefficients over the rational numbers $\mathbb{Q}$.
8. True or False: The volume of the solid bounded by the elliptical paraboloid and the plane is 1.5 cubic units.
9. True or False: The transformation from $\sin x$ to $3 \sin 2x$ does not change the domain, but it scales the range by a factor of three.
10. True or False: Every ring must have a multiplicative identity, which is not true for all rings, such as some rings of functions.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A patient administers a daily dosage of 150 mg of a drug, with 5% remaining in the body just before each dose. Calculate and explain the quantity of the drug present after the third administration.
12. Using the divergence theorem, evaluate the surface integral $\iint_S \vec{F} \cdot d\vec{S}$ for the vector field $\vec{F} = \sin(\pi x)\vec{i} + (zy^3)\vec{j} + (z^2 + 4x)\vec{k}$ over the specified box. Discuss the steps and reasoning behind your solution.

End of Paper